Differentially Private Multiaccuracy Post-processing

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1. Introduction

Differential Privacy (DP) promises that "You will not be affected, adversely or otherwise, by allowing your data to be used in any study or analysis, no matter what other studies, data sets, or information sources, are available." [4] In mathematical terms this means that every randomized-algorithm $\mathcal{M}$ is (ε, δ)-DP if, for all neighboring datasets x and y which differ by a single individual and for all $S \subseteq Range(\mathcal{M})$

$$\Pr[\mathcal{M}(x) \in S] \leq e^{\varepsilon} \Pr[\mathcal{M}(y) \in S] + \delta$$

must hold. [4]

Ultimately, the aim is to protect individuals' privacy such that datasets that may contain sensitive data can be utilized publicly and have important information extracted from them. However, privacy is not the only characteristic that DP should afford to individuals in order to keep the promise, another is fairness.

Balancing privacy and fairness, although not an easy feat, is necessary as often datasets that require applying DP to them also necessitate that the datasets remain representative of the original dataset and thus "fair." For example, when we consider the US census, it is of the utmost that both of these are balanced. Privacy is important as census data can potentially put people at risk of eviction, prosecution, and harassment if their data is leaked. Additionally, fairness must also be prioritized given that the census informs the allocation of trillions of dollars in federal assistance to states and communities. A dataset that has a DP-algorithm applied to it should not apply any additional bias that results in the inequitable distribution of resources. There are many more examples where balancing privacy and fairness is pertinent and this paper seeks to investigate how we can balance that.

Fairness Measurements

Though it is difficult to refute the need for fairness there is much debate as to how to mathematically define fairness. A particularly popular group of definitions are based upon the confusion matrix where we measure fairness based on the rates of: true-positive, false-positives, true-negatives, and false-negatives. For a protected attribute $A \in \{a, b\}$, true label Y, and predicted label $\hat{Y}$ we can write each box of the confusion matrix as follows: [5]
Equal True Positive Rates

$$\Pr(\hat{Y} = 1 \mid Y = 1, A = a) = \Pr(\hat{Y} = 1 \mid Y = 1, A = b)$$

Equal False Positive Rates

$$\Pr(\hat{Y} = 1 \mid Y = 0, A = a) = \Pr(\hat{Y} = 1 \mid Y = 0, A = b)$$

Equal True Negative Rates

$$\Pr(\hat{Y} = 0 \mid Y = 0, A = a) = \Pr(\hat{Y} = 0 \mid Y = 0, A = b)$$

Equal False Negative Rates

$$\Pr(\hat{Y} = 0 \mid Y = 1, A = a) = \Pr(\hat{Y} = 0 \mid Y = 1, A = b)$$

Then, if we put aside error rates, and instead consider risk scores and true outcome probabilities we can also yield the definition of calibration in groups which is defined as, for a score $S \in [0, 1]$, Calibration Within Groups

$$\Pr(Y = 1 \mid S = s, A = a) = \Pr(Y = 1 \mid S = s, A = b) \quad \forall s$$

Kleinberg, Mullainathan, and Raghavan (2016) proves that fairness metrics are mutually incompatible by noting that if the base rates differ

$$\Pr(Y = 1 \mid A = a) \neq \Pr(Y = 1 \mid A = b),$$

and the classifier is not exactly perfect in its accuracy of labeling then no predictor can satisfy all three of: equal false positive rates, equal false negative rates, and calibration. [8] This shows that for the continuous case different fairness metrics can conflict with one another.

Chouldechova's (2017) analysis looked specifically at the COMPAS recidivism risk tool which was a discrete case classifying victims as "high risk" or "low risk". [2] This was well-calibrated across groups, that is defendants with the same predicted risk score had the same reoffense rate, however also resulted in black defendants expericing higher false positive rates and lower false negative rates than their white counterparts. [2] To sum up, both studies showed that choosing to improve one fairness metric may in fact worsen another and this was a mathematical consequence rather than issues with the model or its training.

Differential Privacy vs. Fairness

Not only are fairness metrics at odds with one another, but adding in privacy further complicates the matter. Recent literature shows that DP and exact group fairness cannot, in general, be simultaneously achieved. Cummings et al. (2019) prove that if a classifier is perfectly fair across groups then it cannot achieve high accuracy with a (ε, δ)-DP algorithm. [3] Intuitively this makes sense, fairness requires a model to be responsive to small but systematic group differences whilst DP requires insensitivity for individuals' contributions.

Thus, given that privacy and fairness are inherently incompatible, the authors introduce the concept of approximate fairness. A classifier is α-fair if

$$\left| \Pr(\hat{Y} = 1 \mid Y = y, A = a) - \Pr(\hat{Y} = 1 \mid Y = y, A = b) \right| \leq \alpha.$$

for a parameter α, the classifier is allowed to violate exact fairness by α whilst maintaining (ε, δ)-DP. [3] This provides a compromise where approximate fairness can exist with DP and be

Probably Approximately Correct (PAC) learn accurate. [3]

2. Multiaccuracy Model

In this section we summarize the multiaccuracy framework that our project builds on. Multiaccuracy is a notion of prediction quality that aims to control systematic errors not only on the overall population, but simultaneously across a rich family of (possibly overlapping) subgroups. It provides a way to formalize and improve subgroup fairness for probabilistic classifiers without requiring explicit knowledge of protected attributes at test time. It's also important that the notion of overlapping subgroups is highlighted given that the compounding of many minority subgroups on an individual can majorly affect their result in a classifier.

2.1 Problem Setup

Multiaccuracy is a notion of prediction quality that aims to control systematic errors not only on the overall population, but also across many possibly overlapping subgroups. This is especially useful in fairness settings, where we want to detect and correct patterns such as a classifier consistently underestimating risk for one demographic group while overestimating for another, even when protected attributes are not explicitly available at test time. Essentially, the auditor which we will explain in more detail further on looks to ensure that no group suffers from a systematically incorrect classifier.

Let X denote the input space and let $y : X \to \{0, 1\}$ be the (unknown) ground-truth labeling function. We assume that examples $(x, y(x))$ are drawn from an underlying distribution D over X. A base probabilistic classifier is a function

$$f_0 : X \to [0, 1],$$

where $f_0(x)$ can be interpreted as the model's estimate of $\Pr[y(x) = 1 \mid x]$.

Our goal is to construct a new classifier $f : X \to [0, 1]$ by post-processing f_0 using a small, labeled *audit* dataset sampled from D. We would like f to correct systematic biases of f_0 on many different subpopulations, while preserving or improving its overall accuracy.

To formalize "many different subpopulations," we work with a family of test functions

$$\mathcal{C} \subseteq [-1, 1]^X.$$

Each $c \in \mathcal{C}$ weights examples according to some property of interest. Simple examples include indicator functions of subgroups (e.g., "young", "female", "smoker"), but $\mathcal{C}$ can be much richer and contain, for instance, decision tree predicates or linear threshold functions over features.

2.2 Definition of Multiaccuracy

The central idea of multiaccuracy is to require that the model's prediction errors are uncorrelated with all tests in $\mathcal{C}$. Intuitively, no function in $\mathcal{C}$ should be able to systematically distinguish where the model is over- or under-predicting.

Definition 1 (Differential Privacy)**.** *An algorithm $\mathcal{A}$ with input dataset S and output in a range $\mathcal{R}$ is* (ε, δ)-differentially private *if, for any two datasets S and S' that differ in a single example* *and for any measurable subset $R \subseteq \mathcal{R}$,*

$$\Pr[\mathcal{A}(S) \in R] \ \leq \ e^{\varepsilon} \Pr[\mathcal{A}(S') \in R] + \delta.$$

Definition 2 (Multiaccuracy)**.** *A predictor $f : X \to [0,1]$ is* $(\mathcal{C}, \alpha)$-multiaccurate *with respect to D and y if, for every test function $c \in \mathcal{C}$,*

$$\left|\mathbb{E}_{x \sim D}\left[\, c(x) \left(f(x) - y(x)\right) \,\right]\right| \leq \alpha.$$

The term $f(x) - y(x)$ is the prediction residual. The expectation

$$\mathbb{E}_{x \sim D}\left[c(x)(f(x) - y(x))\right]$$

is the signed average error when examples are weighted by $c(x)$. Requiring this quantity to be small for all $c \in \mathcal{C}$ ensures that:

- For every subgroup whose membership can be represented (even approximately) by some $c \in \mathcal{C}$, the model's average prediction is close to the true average label on that subgroup.
- The model does not systematically overestimate or underestimate the label probability on any such subgroup.

A useful special case is when $\mathcal{C}$ contains indicator functions of subgroups, $c(x) = \mathbf{1}[x \in S]$. Then multiaccuracy implies that for every S of interest,

$$\left|\mathbb{E}_{x \sim D | x \in S}\left[f(x)\right] - \mathbb{E}_{x \sim D | x \in S}\left[y(x)\right]\right|$$

is small. In other words, the predicted positive rate matches the true positive rate on each subgroup S, up to α.

2.3 Multiaccuracy and Subgroup Accuracy

Beyond calibration-style guarantees, multiaccuracy can be connected to clasification accuracy on subgroups. Suppose there is a subgroup S of interest with non-negligible mass under D, and suppose there is a test function $c_S \in \mathcal{C}$ that correlates well with the labels on S (for instance, a weak classifier or a simple regression of y on features restricted to S). Then, if f is $(\mathcal{C}, \alpha)$-multiaccurate, one can show that the classification error of f on S is bounded in terms of α, the approximation error of c_S, and the size of S.

Informally, this means:

- As long as $\mathcal{C}$ is rich enough to detect misprediction patterns on a subgroup, driving the multiaccuracy residuals small forces the model to be reasonably accurate on that subgroup as well.
- Multiaccuracy therefore provides a generic route to improving performance on many overlapping, potentially unlabeled subgroups, without explicitly training a separate classifier per group.

This property is especially attractive in fairness settings, where subgroups may be defined by combinations of attributes and may not all be explicitly enumerated during training.

2.4 Multiaccuracy Boost

In practice, we cannot directly enforce the multiaccuracy constraints over all $c \in \mathcal{C}$, because both D and $\mathcal{C}$ may be large. Instead, [7] proposes an iterative *multiaccuracy boosting* procedure that uses an auditor to search for violations of multiaccuracy and correct them.

At a high level, the algorithm proceeds as follows:

1. **Initialization.** Start from a base classifier f_0.
2. **Auditing step.** On an audit dataset $\{(x_i, y_i)\}$, compute residuals $r_t(x_i) = f_t(x_i) - y_i$. Train an *auditor* $h_t : X \to \mathbb{R}$ to predict these residuals from the features. The auditor's hypothesis class is chosen to approximate functions in $\mathcal{C}$.
3. **Violation check.** If h_t cannot predict the residuals significantly better than chance, this suggests that there is no strong remaining correlation between residuals and any function in the auditor's class, and f_t is approximately $(\mathcal{C}, \alpha)$-multiaccurate. The algorithm stops.
4. **Update step.** Otherwise, the learned auditor h_t certifies a direction in which the model is systematically wrong. The algorithm then updates f_t in the opposite direction of h_t, e.g., using a multiplicative-weights-style update on the predicted probabilities, to obtain f_{t+1} with smaller residuals along that direction.

This boosting-style process is repeated until no significant violation can be found by the auditor. Under mild conditions, the final classifier f_T is guaranteed to be multiaccurate with respect to $\mathcal{C}$.

3. Differentially Private Multiaccuracy

In the non-private multiaccuracy framework (Section 2), the audit dataset S is used to compute residuals, train an auditor to detect systematic errors, and evaluate correlations between the auditor and the model's prediction residuals. Each of these steps directly depends on individual examples from S, so releasing the trained auditors, correlation scores, and/or the final post-processed classifier may reveal sensitive information about the audit data. Our goal in this section is to modify the multiaccuracy boosting procedure so that it satisfies differential privacy with respect to S while retaining its ability to identify and correct subgroup-specific biases.

We thus extend the multiaccuracy framework described in Section 2 to the setting where the audit dataset must remain differentially private. Our goal is to design a post-processing procedure that (i) improves multiaccuracy of a base classifier, and (ii) satisfies differential privacy with respect to the audit examples used in the procedure.

3.1 Differentially Private Multiaccuracy Boost Algorithm

Let $S = \{(x_i, y_i)\}_{i=1}^n$ denote the labeled audit dataset, sampled i.i.d. from the underlying distribution D. We treat S as sensitive and require that our post-processing algorithm be *differentially private* with respect to S.

In our setting, the algorithm $\mathcal{A}$ takes as input the base classifier f_0 and the audit dataset S, and outputs a post-processed classifier f. The differential privacy guarantee requires that the presence or absence of any single individual in S has only a limited effect on the distribution of the final classifier f.

Importantly, we assume that the base classifier f_0 itself is fixed and *non-private*: we do not attempt to protect the data used to train f_0. Differential privacy is imposed only on the audit-driven post-processing, which is the component that directly accesses S. We now describe the differentially private multiaccuracy boosting algorithm, and then justify its privacy guarantees.

In this description, the only steps that access the raw audit dataset S are the residual computation, the DP-SGD auditor training, and the noisy correlation query. Residuals are computed internally and never released; the publicly observable quantities derived from S are the sequence of auditors $\hat{h}_t$, the correlation estimates $\widehat{\Delta}_t$, and the final classifier f_T, all of which are outputs of differentially private mechanisms or their post-processings.

3.2 DP-SGD as a DP-ERM Procedure

We now formalize the privacy guarantee provided by the DP-SGD auditor training routine. We consider the auditor learning problem at a fixed boosting round t as an empirical risk minimization (ERM) task on the residual dataset

$$S_t \;=\; \{(z_i, r_t(z_i))\}_{i=1}^n, \qquad z_i = \phi(x_i), \quad r_t(x_i) = f_t(x_i) - y_i,$$

where ϕ is the feature map and f_t is the current classifier. The auditor is parameterized by $\theta \in \mathbb{R}^d$ with predictions $h_\theta(z)$, and we consider a loss function $\ell(\theta; z, r)$, e.g., squared loss

$$\ell(\theta; z, r) = \big(h_\theta(z) - r\big)^2.$$

The empirical risk on S_t is

$$L(\theta; S_t) \;=\; \frac{1}{n}\sum_{i=1}^n \ell(\theta; z_i, r_t(z_i)).$$

DP-SGD update rule. A single step of (full-batch) DP-SGD at parameters θ computes the per-example gradients

$$g_i(\theta) \;=\; \nabla_\theta \ell(\theta; z_i, r_t(z_i)), \qquad i = 1, \ldots, n,$$

clips each gradient to have norm at most C,

$$\tilde{g}_i(\theta) \;=\; g_i(\theta) \cdot \min\Big(1, \frac{C}{\|g_i(\theta)\|_2}\Big),$$

forms the average

$$q(S_t) \;=\; \frac{1}{n}\sum_{i=1}^n \tilde{g}_i(\theta),$$

and then adds Gaussian noise:

$$\tilde{g} \;=\; q(S_t) + Z, \qquad Z \sim \mathcal{N}(0, \sigma^2 I_d).$$

The parameters are then updated via

$$\theta \leftarrow \theta - \eta \tilde{g}.$$

We denote by $\mathcal{M}_s$ the mechanism that maps the dataset S_t and current parameters θ to the noisy gradient $\tilde{g}$ (and hence to the updated parameters) at step s.

Lemma 1 (Sensitivity of the clipped gradient average)**.** *Consider two neighboring datasets S_t and S_t' that differ in exactly one example. Let $q(S_t)$ and $q(S_t')$ denote the corresponding clipped gradient averages defined above, for a fixed parameter vector θ. Then*

$$\left\| q(S_t) - q(S_t') \right\|_2 \;\leq\; \frac{2C}{n}.$$

In particular, the ℓ_2-sensitivity of the mapping $S_t \mapsto q(S_t)$ is at most $2C/n$.

Proof. Let S_t and S_t' differ in the j-th example. Then for all $i \neq j$ we have $\tilde{g}_i(\theta; S_t) = \tilde{g}_i(\theta; S_t')$, and

$$q(S_t) - q(S_t') \;=\; \frac{1}{n}\big(\tilde{g}_j(\theta; S_t) - \tilde{g}_j(\theta; S_t')\big).$$

By construction of the clipping step, $\|\tilde{g}_j(\theta; S_t)\|_2 \leq C$ and $\|\tilde{g}_j(\theta; S_t')\|_2 \leq C$, so by the triangle inequality

$$\left\|\tilde{g}_j(\theta; S_t) - \tilde{g}_j(\theta; S_t')\right\|_2 \;\leq\; \|\tilde{g}_j(\theta; S_t)\|_2 + \|\tilde{g}_j(\theta; S_t')\|_2 \;\leq\; 2C.$$

Therefore

$$\|q(S_t) - q(S_t')\|_2 \;=\; \frac{1}{n}\left\|\tilde{g}_j(\theta; S_t) - \tilde{g}_j(\theta; S_t')\right\|_2 \;\leq\; \frac{2C}{n},$$

which proves the claim. □

Lemma 2 (One DP-SGD step is differentially private)**.** *Fix a parameter vector θ and a residual dataset S_t. Let $\mathcal{M}_s$ be the mechanism that outputs*

$$\tilde{g}(S_t) \;=\; q(S_t) + Z, \qquad Z \sim \mathcal{N}(0, \sigma^2 I_d),$$

with $q(S_t)$ as above. If

$$\sigma \;\geq\; \frac{2C}{n} \cdot \frac{\sqrt{2\ln(1.25/\delta_s)}}{\varepsilon_s},$$

then $\mathcal{M}_s$ is $(\varepsilon_s, \delta_s)$-differentially private with respect to S_t.

Proof. By Lemma 1, the mapping $S_t \mapsto q(S_t)$ has ℓ_2-sensitivity at most $\Delta = 2C/n$. The mechanism $\mathcal{M}_s$ adds Gaussian noise $Z \sim \mathcal{N}(0, \sigma^2 I_d)$ to $q(S_t)$. By the standard Gaussian mechanism for vector-valued queries, if

$$\sigma \;\geq\; \frac{\Delta\sqrt{2\ln(1.25/\delta_s)}}{\varepsilon_s},$$

then the mechanism $S_t \mapsto q(S_t) + Z$ is $(\varepsilon_s, \delta_s)$-differentially private. Substituting $\Delta = 2C/n$ yields the condition in the statement. Since the subsequent parameter update $\theta \leftarrow \theta - \eta\tilde{g}$ is a deterministic post-processing of $\tilde{g}$, it does not increase the privacy loss. Thus the entire step is $(\varepsilon_s, \delta_s)$-DP with respect to S_t. □

Theorem 1 (DP-SGD auditor training is $(\varepsilon_t^{\mathsf{audit}}, \delta_t^{\mathsf{audit}})$-DP)**.** *Consider running K steps of the DP-SGD procedure described above to train the auditor parameters θ on the residual dataset S_t, starting from some (data-independent) initialization $\theta^{(0)}$. Suppose that at each step $s = 1, \ldots, K$ we choose the noise standard deviation σ_s so that the corresponding mechanism $\mathcal{M}_s$ is $(\varepsilon_s, \delta_s)$-differentially private with respect to S_t. Then the final parameters $\theta^{(K)}$ (and hence the resulting auditor $\hat{h}_t$) are $(\varepsilon_t^{\mathsf{audit}}, \delta_t^{\mathsf{audit}})$-differentially private with*

$$\varepsilon_t^{\mathsf{audit}} = \sum_{s=1}^{K} \varepsilon_s, \qquad \delta_t^{\mathsf{audit}} = \sum_{s=1}^{K} \delta_s.$$

Proof. By Lemma 2, for each s the mechanism $\mathcal{M}_s$ that maps S_t and the current parameters to the noisy gradient (and the updated parameters) is $(\varepsilon_s, \delta_s)$-DP with respect to S_t. The sequence of K update steps is an adaptive composition of these mechanisms: the input to step s may depend on the outcomes of the previous steps, but the dataset S_t itself is fixed.

By the basic composition theorem for differential privacy, the composition of K mechanisms with guarantees $(\varepsilon_s, \delta_s)$ is $(\varepsilon_t^{\mathsf{audit}}, \delta_t^{\mathsf{audit}})$-DP with

$$\varepsilon_t^{\mathsf{audit}} = \sum_{s=1}^{K} \varepsilon_s, \qquad \delta_t^{\mathsf{audit}} = \sum_{s=1}^{K} \delta_s.$$

The final parameter vector $\theta^{(K)}$ is a deterministic function of the sequence of noisy gradients produced by $\mathcal{M}_1, \ldots, \mathcal{M}_K$, and therefore is a post-processing of these differentially private mechanisms. By closure of differential privacy under post-processing, releasing $\theta^{(K)}$ or the induced auditor $\hat{h}_t$ preserves the same $(\varepsilon_t^{\mathsf{audit}}, \delta_t^{\mathsf{audit}})$ guarantee. □

In our implementation, we do not explicitly compute $(\varepsilon_t^{\mathsf{audit}}, \delta_t^{\mathsf{audit}})$ for each round t; instead, we treat the noise multiplier and the number of DP-SGD steps K as proxies for the strength of privacy, following the usual practice in empirical DP-SGD studies. The formal result above shows that, under appropriate calibration of σ_s and using a privacy accountant, the auditor training procedure can be made $(\varepsilon_t^{\mathsf{audit}}, \delta_t^{\mathsf{audit}})$-differentially private with respect to the audit dataset.

3.3 Noisy Correlation Query

We now analyze the privacy of the correlation estimate $\tilde{\Delta}_t$ used in the stopping condition of the boosting procedure. In each round, after training the differentially private auditor $\hat{h}_t$, the algorithm evaluates whether the auditor detects a remaining systematic correlation between the model's residuals and any test in the hypothesis class. Since this correlation is computed directly from the audit dataset S, it must itself be privatized. To privatize this scalar query, we add Gaussian noise. Before doing so, we establish its sensitivity.

Lemma 3 (Sensitivity of the clipped correlation average)**.** *Fix a round t and an auditor $\hat{h}_t$. For each $(x_i, y_i) \in S$ define the per-example score*

$$c_i = \hat{h}_t(x_i)\,(f_t(x_i) - y_i),$$

and let $\tilde{c}_i = \mathrm{clip}(c_i, [-B, B])$. Consider neighboring datasets S and S' that differ in exactly one *example, and let*

$$q_t(S) = \frac{1}{n}\sum_{i=1}^{n} \tilde{c}_i, \qquad q_t(S') = \frac{1}{n}\sum_{i=1}^{n} \tilde{c}'_i.$$

Then

$$\big|q_t(S) - q_t(S')\big| \;\le\; \frac{2B}{n}.$$

In particular, the ℓ_2-sensitivity of the scalar query $S \mapsto q_t(S)$ is at most $2B/n$.

Proof. Let S and S' differ in the j-th example. Then for all $i \neq j$ we have $\tilde{c}_i = \tilde{c}'_i$ and

$$q_t(S) - q_t(S') \;=\; \frac{1}{n}\big(\tilde{c}_j - \tilde{c}'_j\big).$$

By construction, $\tilde{c}_j, \tilde{c}'_j \in [-B, B]$, so

$$|\tilde{c}_j - \tilde{c}'_j| \;\le\; |\tilde{c}_j| + |\tilde{c}'_j| \;\le\; 2B.$$

Therefore

$$|q_t(S) - q_t(S')| \;=\; \frac{1}{n}|\tilde{c}_j - \tilde{c}'_j| \;\le\; \frac{2B}{n},$$

which proves the claim. □

Lemma 4 (Noisy correlation query is $(\varepsilon_t^{\mathsf{corr}}, \delta_t^{\mathsf{corr}})$-DP)**.** *Fix a round t and an auditor $\hat{h}_t$. Let $q_t(S)$ be the clipped correlation average defined above, and consider the Gaussian mechanism*

$$\widehat{\Delta}_t \;=\; q_t(S) + W_t, \qquad W_t \sim \mathcal{N}\big(0, \sigma^2_{\mathrm{corr},t}\big).$$

If the noise standard deviation satisfies

$$\sigma_{\mathrm{corr},t} \;\ge\; \frac{2B}{n} \cdot \frac{\sqrt{2\ln(1.25/\delta_t^{\mathsf{corr}})}}{\varepsilon_t^{\mathsf{corr}}},$$

then the mechanism $S \mapsto \widehat{\Delta}_t$ is $(\varepsilon_t^{\mathsf{corr}}, \delta_t^{\mathsf{corr}})$-differentially private with respect to S.

Proof. By Lemma 3, the scalar query $S \mapsto q_t(S)$ has ℓ_2-sensitivity at most $\Delta = 2B/n$. The mechanism adds Gaussian noise $W_t \sim \mathcal{N}(0, \sigma^2_{\mathrm{corr},t})$ to $q_t(S)$. By the standard Gaussian mechanism for real-valued queries, if

$$\sigma_{\mathrm{corr},t} \;\ge\; \frac{\Delta\sqrt{2\ln(1.25/\delta_t^{\mathsf{corr}})}}{\varepsilon_t^{\mathsf{corr}}},$$

then $S \mapsto q_t(S) + W_t$ is $(\varepsilon_t^{\mathsf{corr}}, \delta_t^{\mathsf{corr}})$-differentially private. Substituting $\Delta = 2B/n$ gives the stated condition. □

3.4 Per-Round and Global Privacy Guarantees

We now combine the two building blocks to show that each boosting round is differentially private, and then use composition across rounds to obtain the global privacy guarantee. Recall that when multiple differentially private mechanisms are applied to the same dataset, their privacy losses accumulate. The *basic composition theorem* states that if two mechanisms are $(\varepsilon_1, \delta_1)$-DP and $(\varepsilon_2, \delta_2)$-DP, then releasing both outputs is $(\varepsilon_1 + \varepsilon_2, \delta_1 + \delta_2)$-DP. We rely on this basic form rather than advanced composition, as our goal is to maintain a clear correspondence between per-round and total privacy budgets.

Proposition 1 (Per-round privacy). *Fix a round t and a current classifier f_t. Suppose we run DP-SGD on the residual dataset S_t to obtain an auditor $\hat{h}_t$ using privacy budget $(\varepsilon_t^{\mathsf{audit}}, \delta_t^{\mathsf{audit}})$ as in Theorem 1, and then compute a noisy correlation estimate $\widehat{\Delta}_t$ using the Gaussian mechanism of Lemma 4 with privacy budget $(\varepsilon_t^{\mathsf{corr}}, \delta_t^{\mathsf{corr}})$. Define the per-round parameters by*

$$\varepsilon_t = \varepsilon_t^{\mathsf{audit}} + \varepsilon_t^{\mathsf{corr}}, \qquad \delta_t = \delta_t^{\mathsf{audit}} + \delta_t^{\mathsf{corr}}.$$

If we choose

$$\varepsilon_t^{\mathsf{audit}} = \varepsilon_t^{\mathsf{corr}} = \frac{\varepsilon_t}{2}, \qquad \delta_t^{\mathsf{audit}} = \delta_t^{\mathsf{corr}} = \frac{\delta_t}{2},$$

then the joint mechanism

$$S \longmapsto (\hat{h}_t, \widehat{\Delta}_t, f_{t+1})$$

is $(\varepsilon_t, \delta_t)$-differentially private with respect to S.

Proof. By Theorem 1, DP-SGD training of the auditor is $(\varepsilon_t^{\mathsf{audit}}, \delta_t^{\mathsf{audit}})$-DP. By Lemma 4, the noisy correlation query is $(\varepsilon_t^{\mathsf{corr}}, \delta_t^{\mathsf{corr}})$-DP. These two mechanisms are applied to the same dataset S in sequence, so by the basic composition theorem for differential privacy, their joint output $(\hat{h}_t, \widehat{\Delta}_t)$ is $(\varepsilon_t, \delta_t)$-differentially private with

$$\varepsilon_t = \varepsilon_t^{\mathsf{audit}} + \varepsilon_t^{\mathsf{corr}}, \qquad \delta_t = \delta_t^{\mathsf{audit}} + \delta_t^{\mathsf{corr}}.$$

Under the specific choice $\varepsilon_t^{\mathsf{audit}} = \varepsilon_t^{\mathsf{corr}} = \varepsilon_t/2$ and $\delta_t^{\mathsf{audit}} = \delta_t^{\mathsf{corr}} = \delta_t/2$, the privacy budget is split evenly between the two sources of privacy loss. This split is chosen for simplicity and symmetry, ensuring each component contributes equally to the total round-level guarantee. We then obtain the stated $(\varepsilon_t, \delta_t)$ guarantee. The update from f_t to f_{t+1} is a deterministic function of $(f_t, \hat{h}_t)$ and does not access S directly. By closure of differential privacy under post-processing, f_{t+1} is also $(\varepsilon_t, \delta_t)$-differentially private with respect to S.

□

Theorem 2 (Global privacy of DP multiaccuracy boosting). *Suppose Algorithm 1 is run with per-round privacy budgets $(\varepsilon_t, \delta_t)$ satisfying $\sum_{t=0}^{T-1} \varepsilon_t \le \varepsilon$ and $\sum_{t=0}^{T-1} \delta_t \le \delta$, and with the fixed split*

$$\varepsilon_t^{\mathsf{audit}} = \varepsilon_t^{\mathsf{corr}} = \frac{\varepsilon_t}{2}, \qquad \delta_t^{\mathsf{audit}} = \delta_t^{\mathsf{corr}} = \frac{\delta_t}{2}$$

for auditor training and correlation estimation. Then the overall algorithm is (ε, δ)-differentially private with respect to the audit dataset S.

Proof. By Proposition 1, the mechanism implemented in round t that maps S to $(\hat{h}_t, \widehat{\Delta}_t, f_{t+1})$ is $(\varepsilon_t, \delta_t)$-differentially private with respect to S. The full boosting procedure applies these per-round mechanisms sequentially for $t = 0, \dots, T-1$, always on the same underlying dataset S. By the basic composition theorem for differential privacy, the composition of these T mechanisms is (ε, δ)-DP with

$$\varepsilon \;\ge\; \sum_{t=0}^{T-1} \varepsilon_t, \qquad \delta \;\ge\; \sum_{t=0}^{T-1} \delta_t.$$

Since we choose the per-round budgets so that $\sum_{t=0}^{T-1} \varepsilon_t \leq \varepsilon$ and $\sum_{t=0}^{T-1} \delta_t \leq \delta$, the overall procedure is indeed (ε, δ)-differentially private with respect to S. □

Fairness-Utility Trade-offs The differentially private multiaccuracy procedure inherits the fairness guarantees of the non-private multiaccuracy framework, up to additional error terms introduced by the DP noise. In particular, as long as the auditor hypothesis class $\mathcal{H}$ is sufficiently expressive to detect subgroup-specific misprediction patterns, driving the (noisy) multiaccuracy residuals towards zero encourages good performance on a wide range of subgroups.

At the same time, the noise added to ensure differential privacy can degrade both overall accuracy and subgroup performance. The severity of this degradation depends on the privacy budget (ε, δ), the size of the audit dataset, the number of boosting rounds T, and the complexity of the auditor class $\mathcal{H}$. A key empirical question in our experiments is therefore to quantify the trade-off between privacy, multiaccuracy (and fairness), and predictive performance, as we vary the strength of the differential privacy constraints.

Algorithm 1 Differentially Private Multiaccuracy Boosting

Require: Base classifier f_0, audit dataset $S = \{(x_i, y_i)\}_{i=1}^n$, feature map ϕ, hypothesis class $\mathcal{H}$, learning rate η, number of rounds T, global privacy budget (ε, δ), clipping norms C (gradients), B (correlations), number of DP–SGD steps K

Choose per-round budgets $(\varepsilon_t, \delta_t)$ s.t. $\sum_{t=0}^{T-1} \varepsilon_t \leq \varepsilon$ and $\sum_{t=0}^{T-1} \delta_t \leq \delta$
Initialize f_0
for $t = 0, 1, \ldots, T-1$ **do**
  Compute residuals $r_t(x_i) = f_t(x_i) - y_i$ for all $(x_i, y_i) \in S$ (internal, not released)
  Form residual dataset $S_t = \{(z_i, r_t(x_i))\}_{i=1}^n$ where $z_i = \phi(x_i)$
  DP–SGD auditor training using privacy $(\varepsilon_t/2, \delta_t/2)$
  Set per-step privacy parameters $\varepsilon_{t,s} = \varepsilon_t/(2K)$, $\delta_{t,s} = \delta_t/(2K)$
  Initialize auditor parameters $\theta_t^{(0)}$ (independent of S)
  for $s = 1, \ldots, K$ **do**
    For each i, compute gradient $g_i^{(s)} = \nabla_\theta \ell(\theta_t^{(s-1)}; z_i, r_t(x_i))$
    Clip: $\tilde{g}_i^{(s)} = g_i^{(s)} \cdot \min(1, C/\|g_i^{(s)}\|_2)$
    Average: $q_t^{(s)}(S_t) = \frac{1}{n} \sum_{i=1}^n \tilde{g}_i^{(s)}$
    Choose $\sigma_{t,s} = \frac{2C}{n} \cdot \frac{\sqrt{2 \ln(1.25/\delta_{t,s})}}{\varepsilon_{t,s}}$
    Sample $Z_{t,s} \sim \mathcal{N}(0, \sigma_{t,s}^2 I)$ and set noisy gradient $\tilde{g}^{(s)} = q_t^{(s)}(S_t) + Z_{t,s}$
    Update $\theta_t^{(s)} = \theta_t^{(s-1)} - \eta_{\text{audit}} \tilde{g}^{(s)}$
  end for
  Let $\hat{h}_t$ be the auditor defined by $\theta_t^{(K)}$
  DP correlation estimate using privacy $(\varepsilon_t/2, \delta_t/2)$
  For each $(x_i, y_i) \in S$ compute $c_i = \hat{h}_t(x_i)\,(f_t(x_i) - y_i)$
  Clip: $\tilde{c}_i = \text{clip}(c_i, [-B, B])$
  Average: $q_t(S) = \frac{1}{n} \sum_{i=1}^n \tilde{c}_i$
  Set $\varepsilon_t^{\text{corr}} = \varepsilon_t/2$, $\delta_t^{\text{corr}} = \delta_t/2$
  Choose $\sigma_{\text{corr},t} = \frac{2B}{n} \cdot \frac{\sqrt{2 \ln(1.25/\delta_t^{\text{corr}})}}{\varepsilon_t^{\text{corr}}}$
  Sample $W_t \sim \mathcal{N}(0, \sigma_{\text{corr},t}^2)$ and set $\widehat{\Delta}_t = q_t(S) + W_t$
  if $|\widehat{\Delta}_t|$ is below a pre-specified threshold **then**
    break (no significant multiaccuracy violation detected)
  end if
  Multiaccuracy update (post-processing)
  Update f_t to f_{t+1} by $f_{t+1}(x) \propto f_t(x) \cdot \exp(-\eta \hat{h}_t(x))$ and project to $[0, 1]$
end for
return f_T

4. Data Applications

The original paper looked to recreate the environment which "Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification" operated under where they studied classifiers of faces on different races and genders. [1] Multiaccuracy Boost was audited using data from "Labelled Faces in the Wild (LFW) Dataset" which contains 13,233 images of 5,749 people. [6] We will be using the same dataset to analyze the effectiveness of the ammended Multiaccuracy Boost algorithm as a direct comparable to see how effective the algorithm is on the same data but with the addition of DP. We use the preprocessed representation from the original multiaccuracy codebase, in which each face is embedded into a 100-dimensional latent vector $X \in \mathbb{R}^{100}$ and annotated with 73 binary facial attributes (e.g., `Male`, `Asian`, `Smiling`).

In all experiments reported below we fix the prediction target to be the attribute $\texttt{Frowning}$, i.e., $Y = \mathbf{1}\{\texttt{Frowning} = 1\}$, and treat race indicators as protected attributes.

Concretely, we consider four protected attributes, $A \in \{\texttt{Indian}, \texttt{Asian}, \texttt{White}, \texttt{Black}\}$, and for each choice of A we create a binary group label $G = \mathbf{1}\{A = 1\}$. The marginal positive rate for the target label is approximately $\Pr(Y = 1) \approx 0.58$, while the protected groups are imbalanced: only about 2.3% of examples are labeled $\texttt{Indian}$, increasing to 7% $\texttt{Black}$, 7.7% $\texttt{Asian}$, and 74.8% $\texttt{White}$.

We randomly split the dataset into a training set, an audit set, and a test set with proportions 0.6/0.2/0.2, stratifying on Y so that the label distribution is comparable across splits. On the training split we fit a logistic regression classifier

$$f_0(x) = \Pr_{\theta}(Y = 1 \mid x)$$

with the standard ℓ_2-regularized logistic loss. This serves as the non-private base model. We then apply two post-processing procedures to this base classifier:

- **Non-private multiaccuracy (MA).** We run the multiplicative-weights multiaccuracy boost of Section 2 using a linear regression auditor trained on the audit split. At each round t the auditor is fit to the residuals $f_t(x) - y$ on the audit set, and we update the logits of f_t in the negative direction of the auditor. We use a learning rate $\eta = 0.1$, at most $T = 10$ boosting rounds, and a stopping threshold of $|\Delta_t| < 0.005$ on the exact correlation
$$\Delta_t = \mathbb{E}_{(x,y)\sim S_{\text{audit}}}\big[h_t(x)\,(f_t(x) - y)\big].$$

- **Differentially private multiaccuracy (DP-MA).** We run the differentially private boosting algorithm of Section 3 with the same base model and audit split. The auditor is trained using DP-SGD with per-round privacy parameters $(\varepsilon_{\text{round}}, \delta_{\text{round}}) = (1, 10^{-2})$, gradient clipping norm $C = 1$, correlation clipping bound $B = 1$, learning rate 0.05, 200 gradient steps, and minibatch size 256. Each boosting round splits its $(\varepsilon_{\text{round}}, \delta_{\text{round}})$ evenly between the DP-SGD auditor training and the noisy correlation query. We again use at most $T = 10$ rounds and a stopping threshold $|\widehat{\Delta}_t| < 0.005$ on the noisy correlation estimate.

For each protected attribute $A \in \{\texttt{Indian}, \texttt{Asian}, \texttt{White}, \texttt{Black}\}$ and each of the three models (Base, MA, DP-MA), we evaluate on the held-out test set the overall accuracy and the group-specific accuracies on $G = 0$ and $G = 1$. As a simple group fairness metric we report the absolute difference

$$\text{gap}(A) \;=\; \big|\text{Acc}(G=0) - \text{Acc}(G=1)\big|$$

between the two groups.

4.1 Results on race subgroups

Table 1 summarizes the performance of the base model, non-private multiaccuracy (MA), and differentially private multiaccuracy (DP-MA) for the target attribute $\texttt{Frowning}$ and race-based protected attributes. Overall, the non-private multiaccuracy procedure produces only small perturbations to the base classifier, whereas the DP-MA procedure substantially degrades accuracy while sometimes reducing, and sometimes worsening, group accuracy gaps.

Protected	Model	Overall acc	Acc(G=0)	Acc(G=1)	$\lvert \mathrm{Acc}_0 - \mathrm{Acc}_1 \rvert$
`Indian`	Base	0.713	0.714	0.643	0.071
	MA	0.711	0.713	0.625	0.088
	DP-MA	0.617	0.617	0.607	0.010
`Asian`	Base	0.713	0.713	0.710	0.003
	MA	0.711	0.712	0.694	0.018
	DP-MA	0.617	0.617	0.612	0.005
`White`	Base	0.713	0.725	0.709	0.016
	MA	0.711	0.722	0.708	0.014
	DP-MA	0.617	0.665	0.602	0.063
`Black`	Base	0.713	0.711	0.735	0.024
	MA	0.711	0.709	0.735	0.026
	DP-MA	0.617	0.615	0.645	0.030

Table 1: Test accuracy of the base logistic regression classifier, non-private multiaccuracy (MA), and differentially private multiaccuracy (DP-MA) for the target attribute `Frowning` and four race-based protected attributes.

For the `Indian` subgroup, which is the smallest and most imbalanced group (only 2.3% of the dataset), the base classifier exhibits a noticeable accuracy gap of 0.071 between non-Indian and Indian faces. The non-private multiaccuracy procedure slightly reduces overall accuracy from 0.713 to 0.711 and actually increases the group gap to 0.088. In contrast, the DP-MA procedure yields a much more balanced classifier, the gap between $G = 0$ and $G = 1$ shrinks to 0.010, so that the accuracies of the two groups become nearly equal (0.617 vs. 0.607). However, this parity comes at a substantial cost in utility, the overall accuracy drops by almost ten percentage points, from 0.713 to 0.617.

For the other protected attributes (`Asian`, `White`, and `Black`), the base model already achieves relatively small group accuracy gaps (between 0.003 and 0.024), so there is limited headroom for further fairness improvement. Non-private multiaccuracy has only a minor effect in these cases: overall accuracy decreases slightly (from 0.713 to 0.711) and the group gaps change by at most a few percentage points, sometimes improving and sometimes worsening, which is consistent with the view that the base classifier is already close to multiaccurate with respect to the chosen linear auditor class.

Overall, these results provide a concrete empirical instance of multiaccuracy-style post-processing that is provably (ε, δ)-differentially private while still delivering non-trivial accuracy and approximately balanced error across several race subgroups. This aligns with the theoretical incompatibility between exact group fairness and differential privacy [3], but also shows that, in practice, one can navigate this trade-off: non-private multiaccuracy has a benign effect when the base model is already close to multiaccurate, and the DP-MA procedure demonstrates that meaningful subgroup fairness improvements are achievable under explicit differential privacy constraints, at the cost of a controlled drop in overall utility.

5. Conclusion

In conclusion, we investigated whether the algorithm "Multiaccuracy Boost" can be made differentially private whilst still retaining the accuracy and fairness that [7] ensured. Using Multiaccuracy Boost as a basis we went on and developed an auditor that was trained with DP-SGD and a stopping criteria based on the Gaussian-noise correlation estimate. Subsequently, we proceeded to prove that our modified algorithm is (ε, δ)-differentially private with respect to S, our raw audit dataset. Our data analysis showed results consistent with prior literature which demonstrated that ensuring all of fairness, accuracy, and privacy was impossible. Our DP-MA procedure substantially improved fairness across the groups and provided DP however came at the cost of accuracy. In terms of future directions in which we could take our project a priority would be in in looking more closely at how to optimize the balance between all three characteristics given that there is an inherent trade off. Furthermore, we'd want to extend onto different datasets beyond the LFW one used in this project to test the ranges of the ideal parameters for different datasets. Ultimately, given the importance of balancing fairness, accuracy, and privacy more work can always be done but we are proud of the work we've added to show that there can be compromise without total failure in any of the areas.

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